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AND ARTS
–
Fakultät IV
Wirtschaft und
Informatik



Fortgeschrittene Aspekte verteilter Systeme

Cloud-Native Computing

Hans Grönniger

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Organization



Organization

- Lectures
  - Cloud (native) computing and software engineering for the cloud
  - Some theory but otherwise very “hands-on”
  - The accompanying project is a programming exercise mostly
- Exercises
  - Development of a web application first as a monolith, later as a cloud-native app

- See Moodle course for all slides, documents, discussions, announcements
  - https://moodle.hs-hannover.de/course/view.php?id=32888



Organization

- Modulprüfung / Exam
  - 55 points in written test (during normal exam period at end of the semester / start of next)
  - 45 points in project tasks
  - 3 presentation slots, attendance is required, active discussion is required
  - You need to obtain at least 50% in the written test AND 50% in the project tasks (overall) to pass the module



Learning Goals

Upon completion, students will be able to:

- model a domain with DDD,
- design and implement a modular monolith,
- Containerize applications and deploy them automatically,
- Evaluate reasons for and against service extraction,
- use synchronous and asynchronous communication,
- distinguish eventual consistency from transactional consistency,
- apply Saga, Retry and other patterns correctly,
- (examine a distributed system with logs, metrics, and traces,)
- reflect on the technical, business and economic consequences of a microservice architecture.



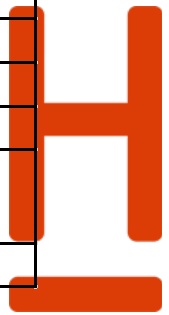
What are your expectations?

- Discuss 2-3 minutes with your neighbor



Content & Timing (prelim)

#	lectures		exercises	
1	29.09.26	Intro, Definition, Market, ShareWise user journey	05.10.26	a) Compare NIST to ISO/IEC 22123(esp. the service models part) b) Familiarize with ShareWise User Journey, work on a more complete example
2	06.10.26	DDD foundation	12.10.26	Elaborate on ShareWise 1) Subdomains, 2) Bounded Contexts, 3) Context Map, 4) Ubiquitous Language
3	13.10.26	Spring Boot, Nextjs	19.10.26	Elaborate on ShareWise 1) Entities, Value Objects 2) Aggregates, 3) Domain Events, 4) (Hexagonal) Architecture
4	20.10.26	Presentations: modular architecture of ShareWise	26.10.26	Implementation Modulith (backend, frontend) - using AI possible
5	27.10.26	Cloud economy, Cloud-native computing, Virtualization	02.11.26	Implementation Modulith (backend, frontend) - using AI possible
6	03.11.26	Container	09.11.26	Create containers and execute locally
7	10.11.26	DevOps & CI/CD	16.11.26	Create pipelines (to build)
8	17.11.26	Presentations: monolith implementation, limitations and change drivers (more users, more scaling, …)	23.11.26	Deployment to eduDSCloud / proxmox (adapt pipeline)
9	24.11.26	IaaS (Azure)	30.11.26	Factor out services, communicate via messages
10	01.12.26	strangler fig, loose coupling, messaging, rabbitmq, outbox	07.12.26	Implement patterns
11	08.12.26	API-Gateway, Eventual Consistency, Saga, Resilience Pattern	14.12.26	Implement patterns
12	15.12.26	Presentations: microservices architecture and pattern implementation	04.01.27	Rework if needed
13	05.01.27	buffer		



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Cloud Computing

History, Definition

The interesting thing about cloud computing is that we've redefined cloud computing to include everything that we already do.

– Richard Stallman



IT Infrastructure

Basis for all services and applications

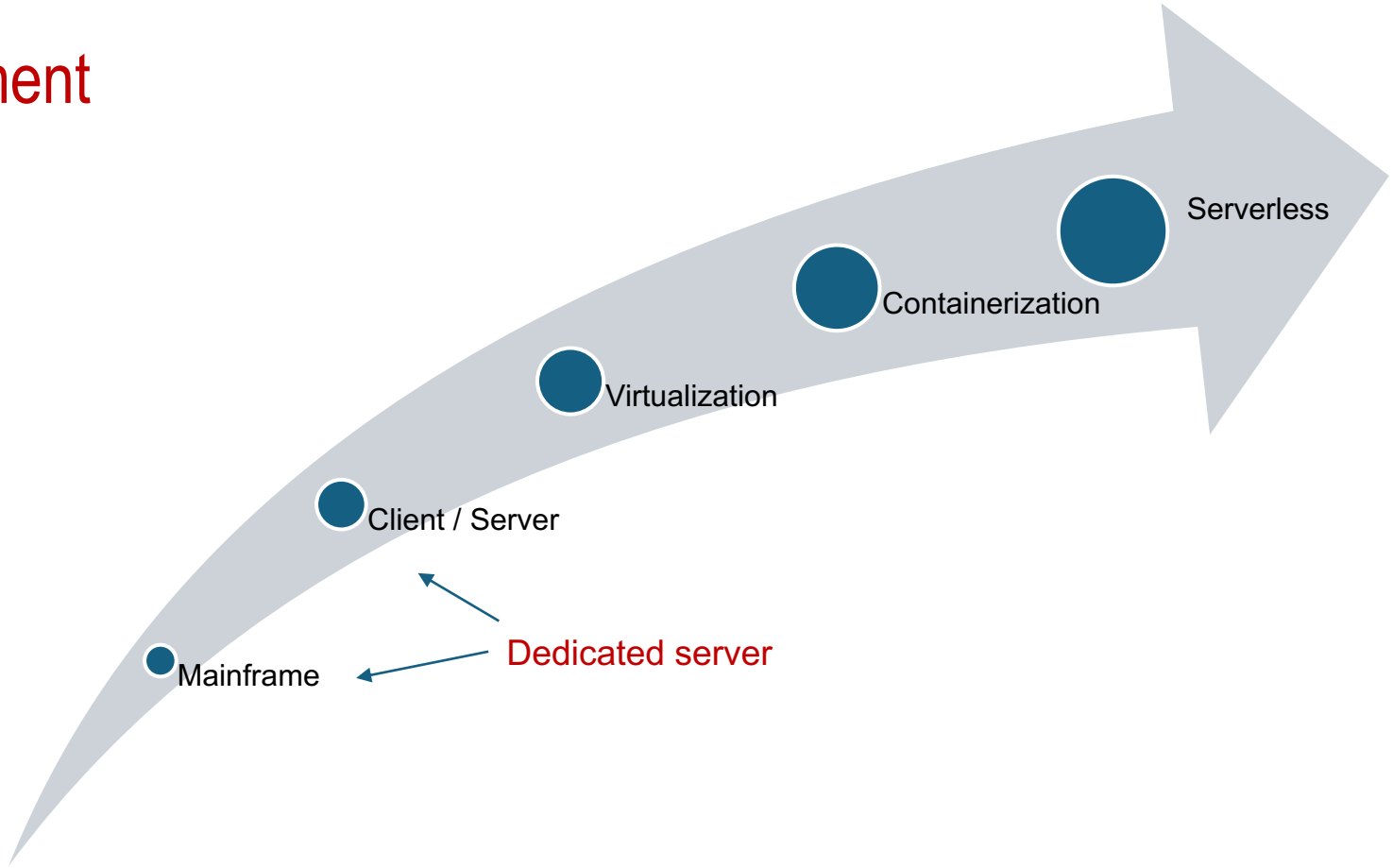
Components

- Network incl. connection to external networks (Internet) & security such as firewalls
- Hardware / machines (all IT components such as servers, routers, PCs, ... but possibly also data center buildings with cooling, redundant power supply, etc.)
- Software (standard software, operating systems, etc.)

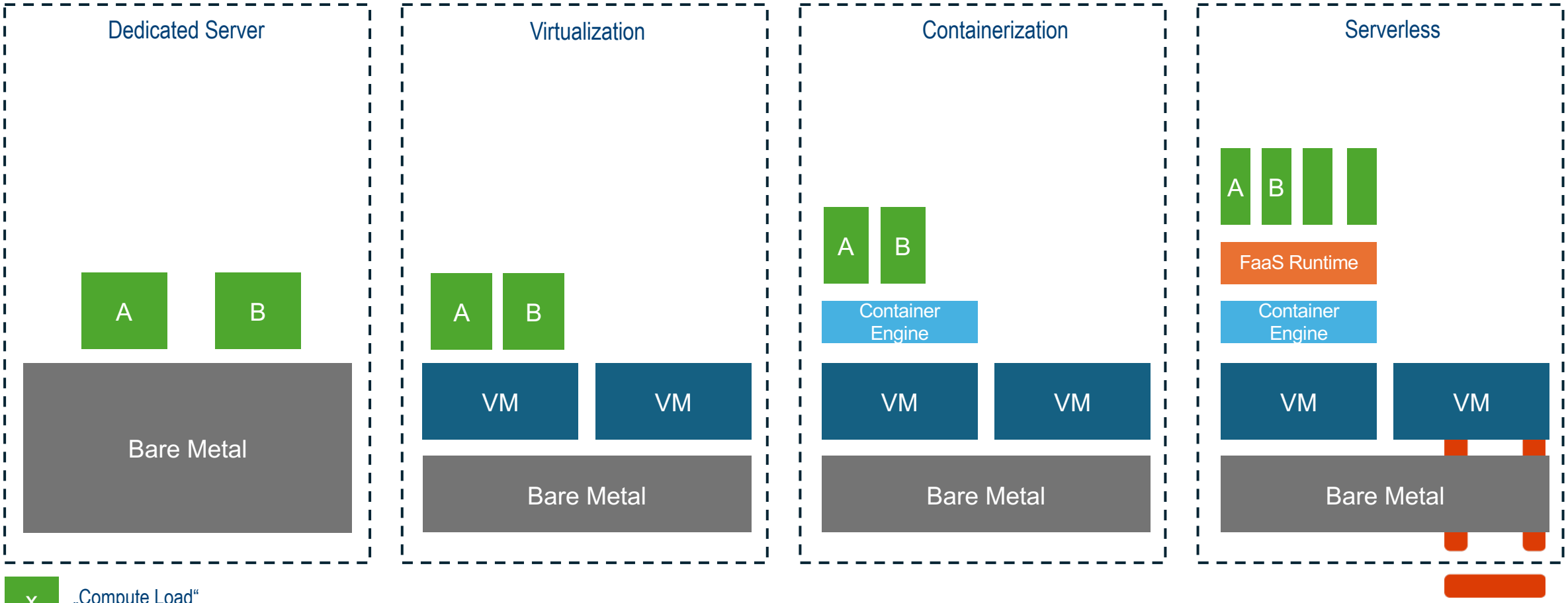


IT Infrastructure

- Development



Deployment Units are getting smaller



IT Infrastructure

Traditional IT infrastructure

- Own data centers (buildings)
- Own hardware
- Own operating staff

Intermediate step: Virtualization

Infrastructure-as-a-Service

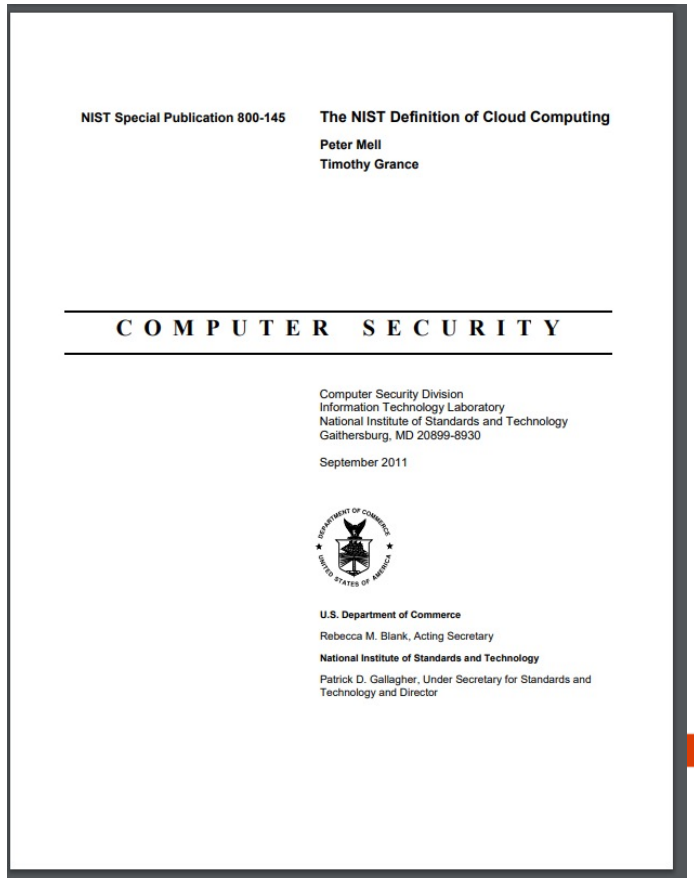
- Rental model
- Public cloud





Cloud Computing

- Definition
  - Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.
  - This cloud model is composed of five essential characteristics, three service models, and four deployment models.


NIST Special Publication 800-145

The NIST Definition of Cloud Computing

Peter Mell
Timothy Grance

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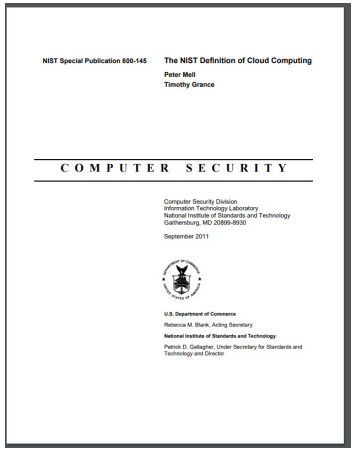
September 2011

U.S. Department of Commerce
Rebecca M. Blank, Acting Secretary

National Institute of Standards and Technology
Patrick D. Gallagher, Under Secretary for Standards and Technology and Director

https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-145.pdf

Cloud Computing


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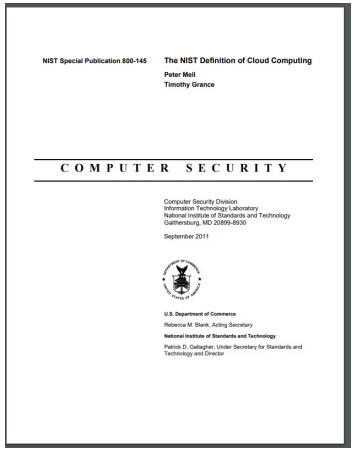
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- Essential characteristics
  1) On-demand self-service. A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
  2) Broad network access. Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).
  3) Resource pooling. The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter). Examples of resources include storage, processing, memory, and network bandwidth.
  4) Rapid elasticity. Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.
  5) Measured service. Cloud systems automatically control and optimize resource use by leveraging a metering capability (e.g. pay-per-use) at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.



Cloud Computing


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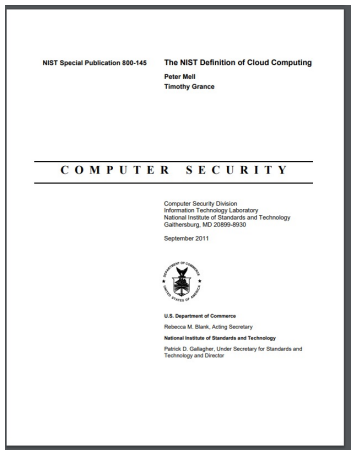
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- Service Models
  1) Software as a Service (SaaS). The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through either a thin client interface, such as a web browser (e.g., web-based email), or a program interface. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.
  2) Platform as a Service (PaaS). The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.
  3) Infrastructure as a Service (IaaS). The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls).

Cloud Computing


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- Deployment Models
  1) Private cloud. The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.
  2) Community cloud. The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.
  3) Public cloud. The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.
  4) Hybrid cloud. The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

Cloud Computing

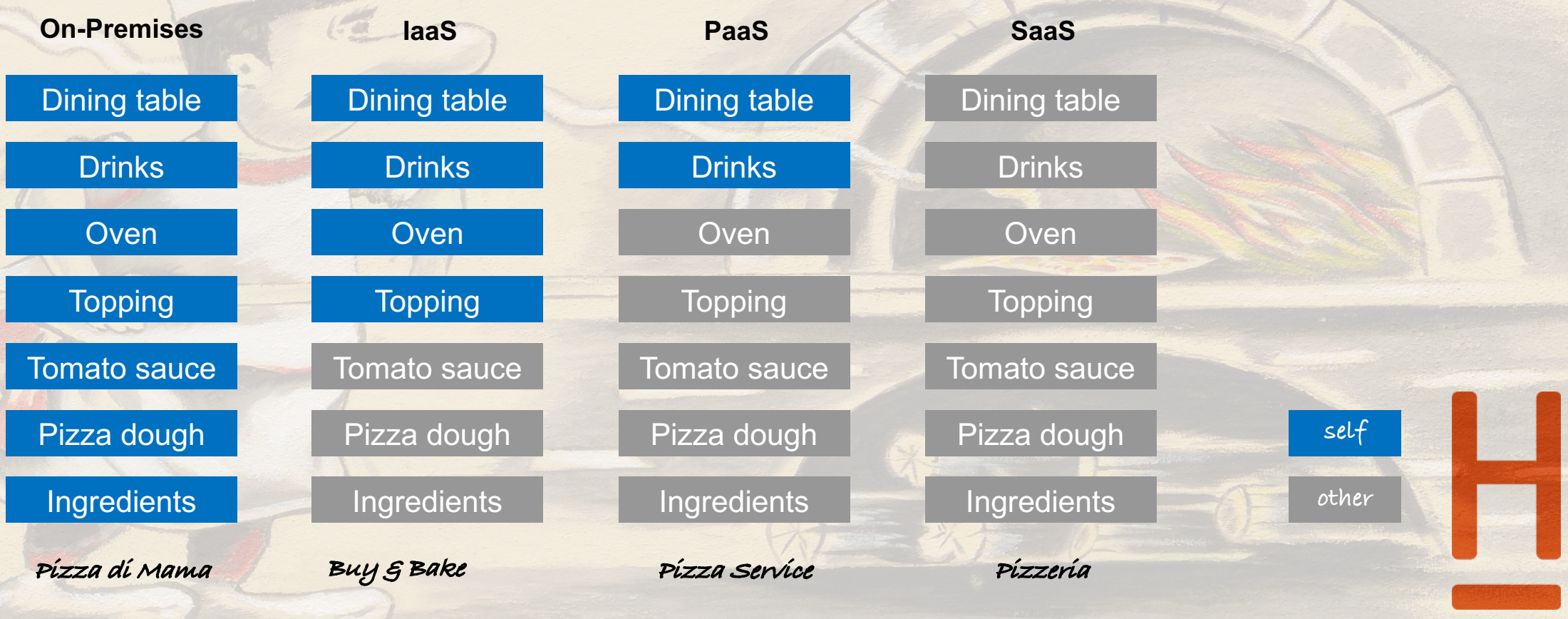
On-Premises	Infrastructure (as a Service)	Platform (as a Service)	Software (as a Service)
“own it”	“host on it“	“build on it“	“consume it“
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Middleware	Middleware	Middleware	Middleware
Runtime	Runtime	Runtime	Runtime
O/S	O/S	O/S	O/S
Virtualization	Virtualization	Virtualization	Virtualization
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

other

self



Pizza-as-a-Service



Quelle: Nane Kratzke, Cloud-native Computing, 2022

Cloud Computing – Chances & Risks

- Discuss 2-3 minutes with you neighbor



Cloud Computing – Chances

- Cost: Companies don't need to make large capital investments in hardware and software.
- Scalability: Cloud resources can be easily scaled as needed, both vertically and horizontally.
- Performance: Major cloud service providers operate data centers around the world, allowing resources to be provisioned quickly and efficiently.
- Speed and agility: With just a few clicks, new resources can be provisioned or released in the cloud. This accelerates the innovation process and provides a competitive advantage.
- Security: Many cloud providers offer a set of policies, technologies, and controls that can improve overall security by protecting your data, applications, and infrastructure from potential threats.
- Recoverability and backup: The cloud enables easier data backup, disaster recovery, and business continuity compared to on-premises solutions.
- Automation: Routine tasks such as backup, software updates, and monitoring can be automated, allowing IT departments to focus on more strategic goals.

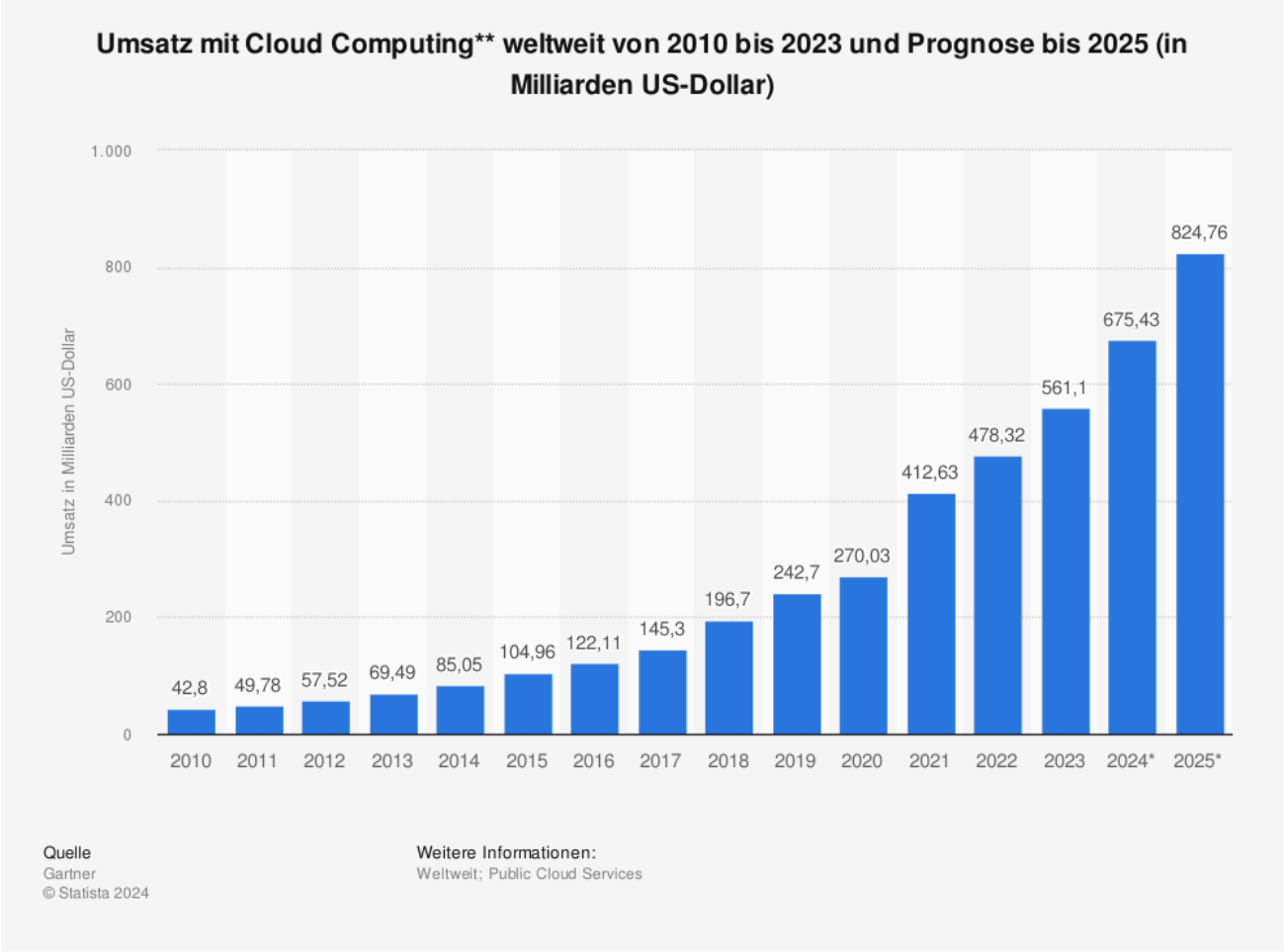


Cloud Computing – Risks

- Cost: Although cloud computing makes it easy to get started, the cost can be high in the long run, especially if services aren't properly managed or optimized.
- Vendor lock-in: Once systems are set up in a dedicated cloud environment, switching to another vendor can be complex and expensive.
- Data protection and compliance: Especially in regions with strict data protection laws, such as the EU, storing data in cloud servers located in other jurisdictions can be problematic.
- Security risks: Although many cloud providers offer high security standards, storing data and applications in the cloud remains a potential risk for data leaks, hacks, and other types of cyberattacks.
- Performance: Since cloud services are accessible via the internet, performance is highly dependent on the quality of the internet connection. In addition, latency and bandwidth can affect performance.
- Data transfer costs: Moving large amounts of data in and out of the cloud can be time-consuming and potentially expensive, depending on the cloud provider's bandwidth and cost structures.
- Loss of control: Outsourcing IT infrastructure to a third party gives away a certain degree of control

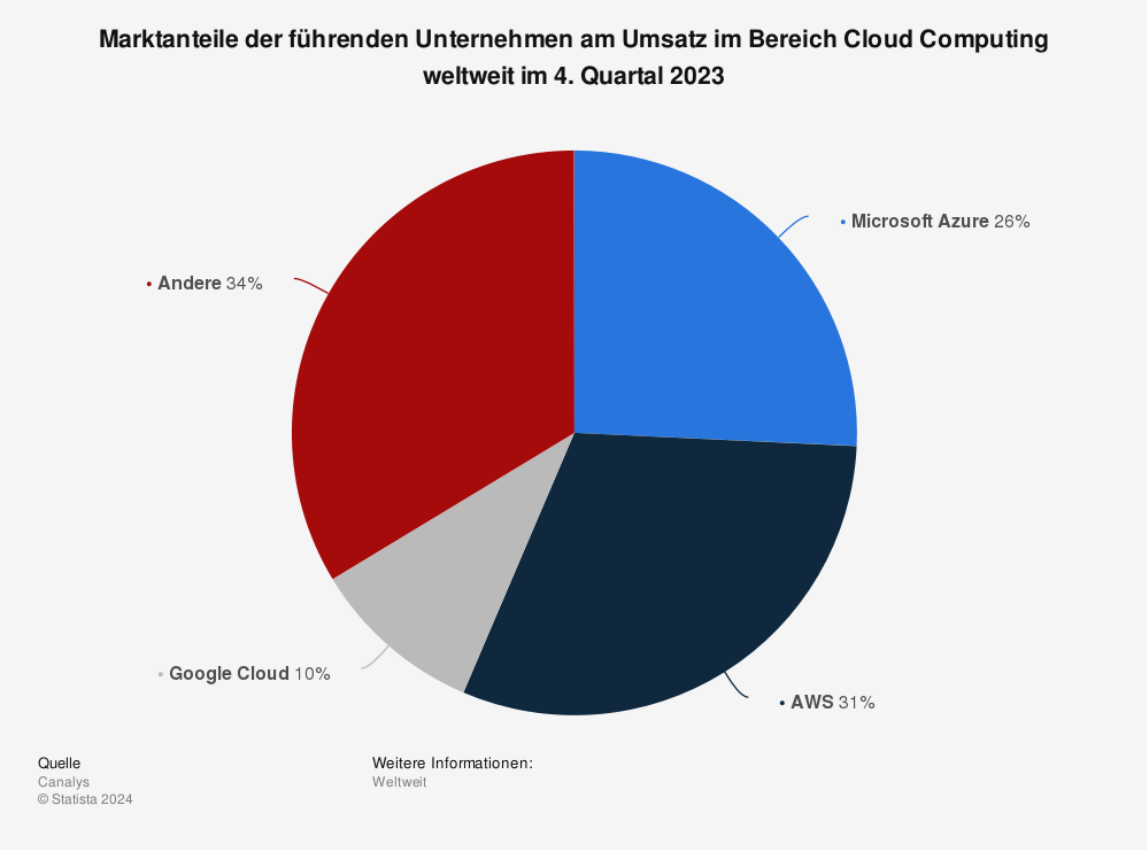


Cloud Computing – Market Figures



https://de.statista.com/themen/562/cloud-computing/#dossierKeyfigures

Cloud Computing – Market Figures

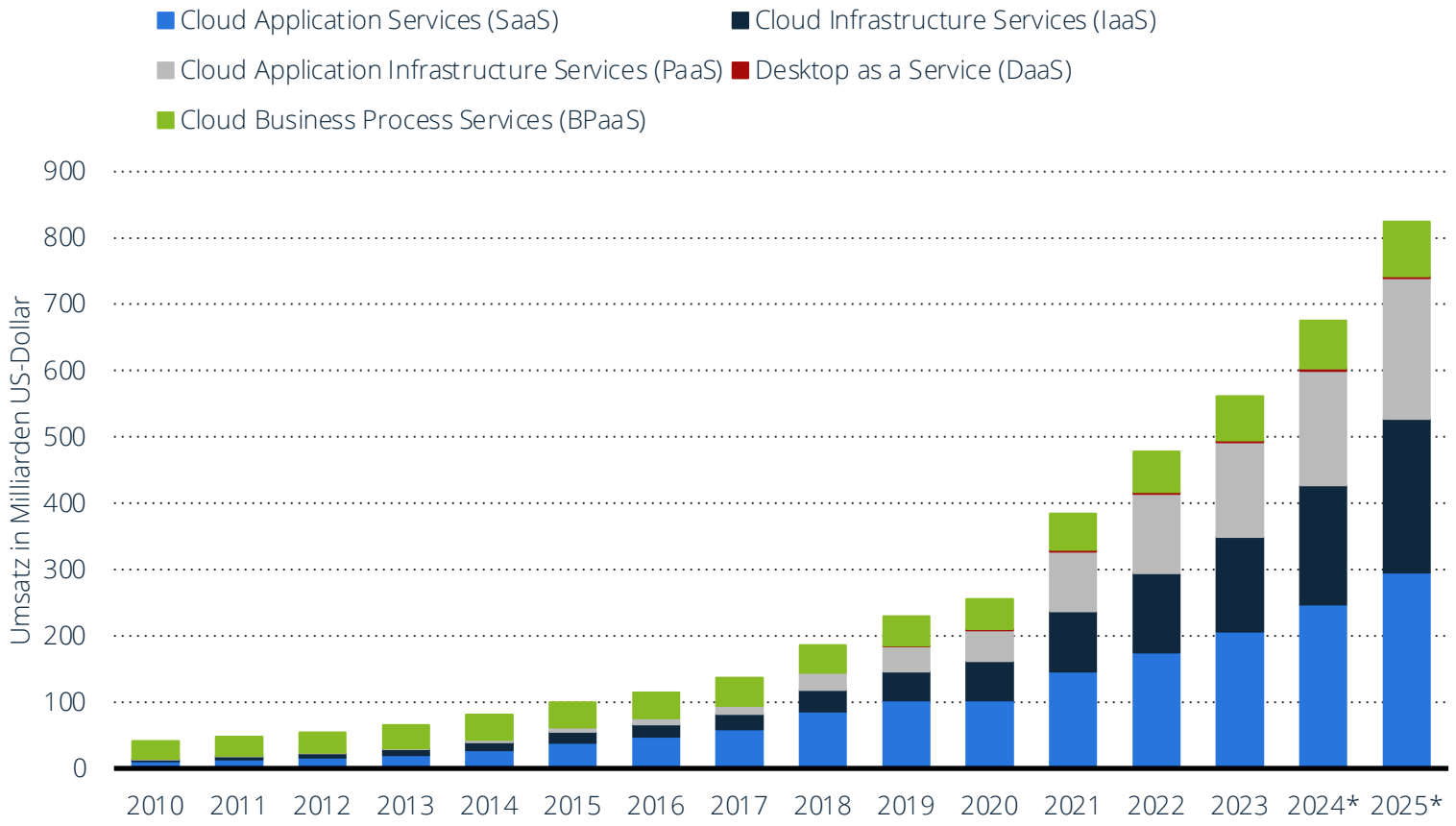


https://de.statista.com/themen/562/cloud-computing/#dossierKeyfigures

BIG-FIVE:

- Amazon
- Microsoft
- Google
- Alibaba
- IBM

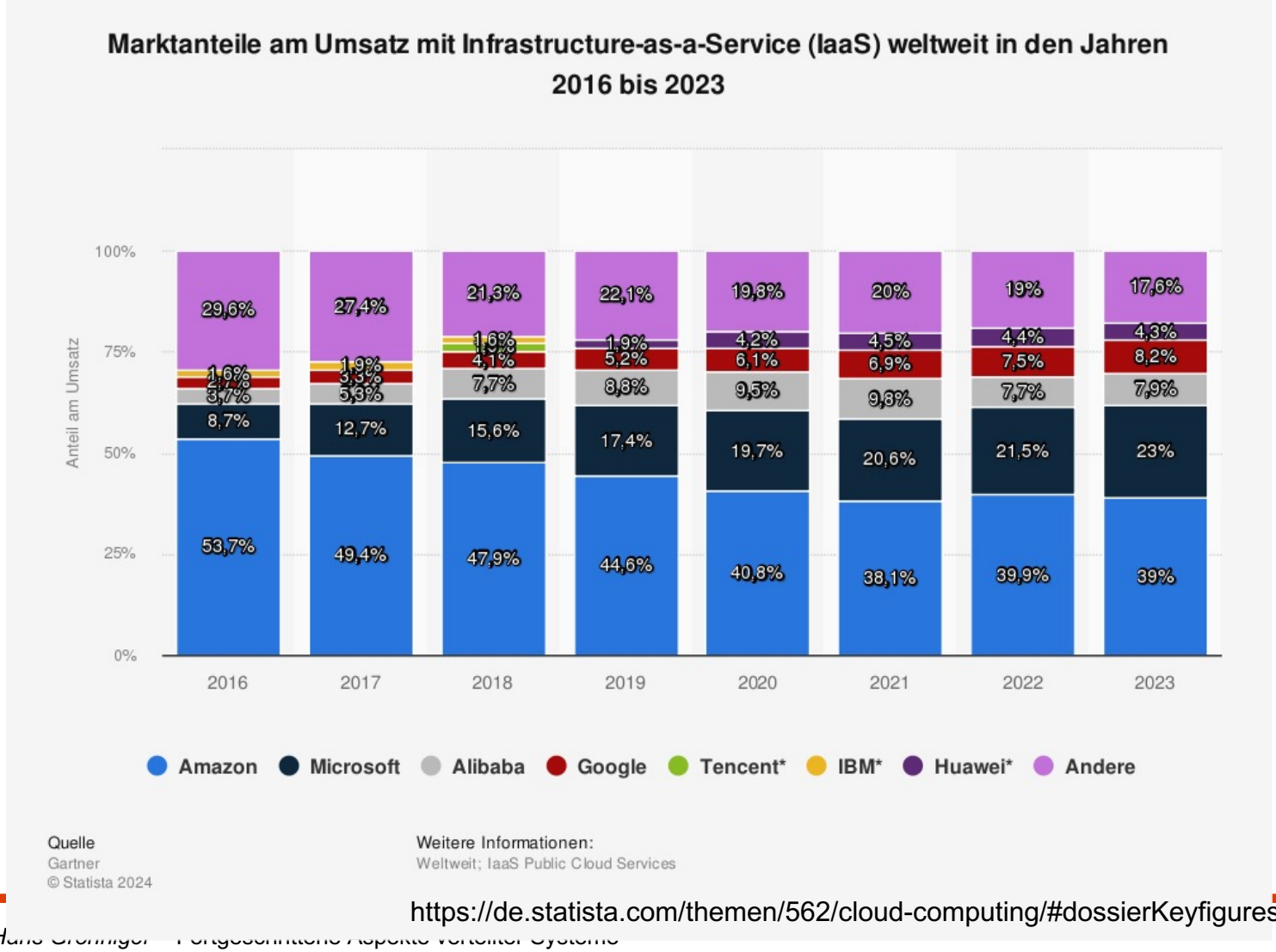
Cloud Computing – Market Figures



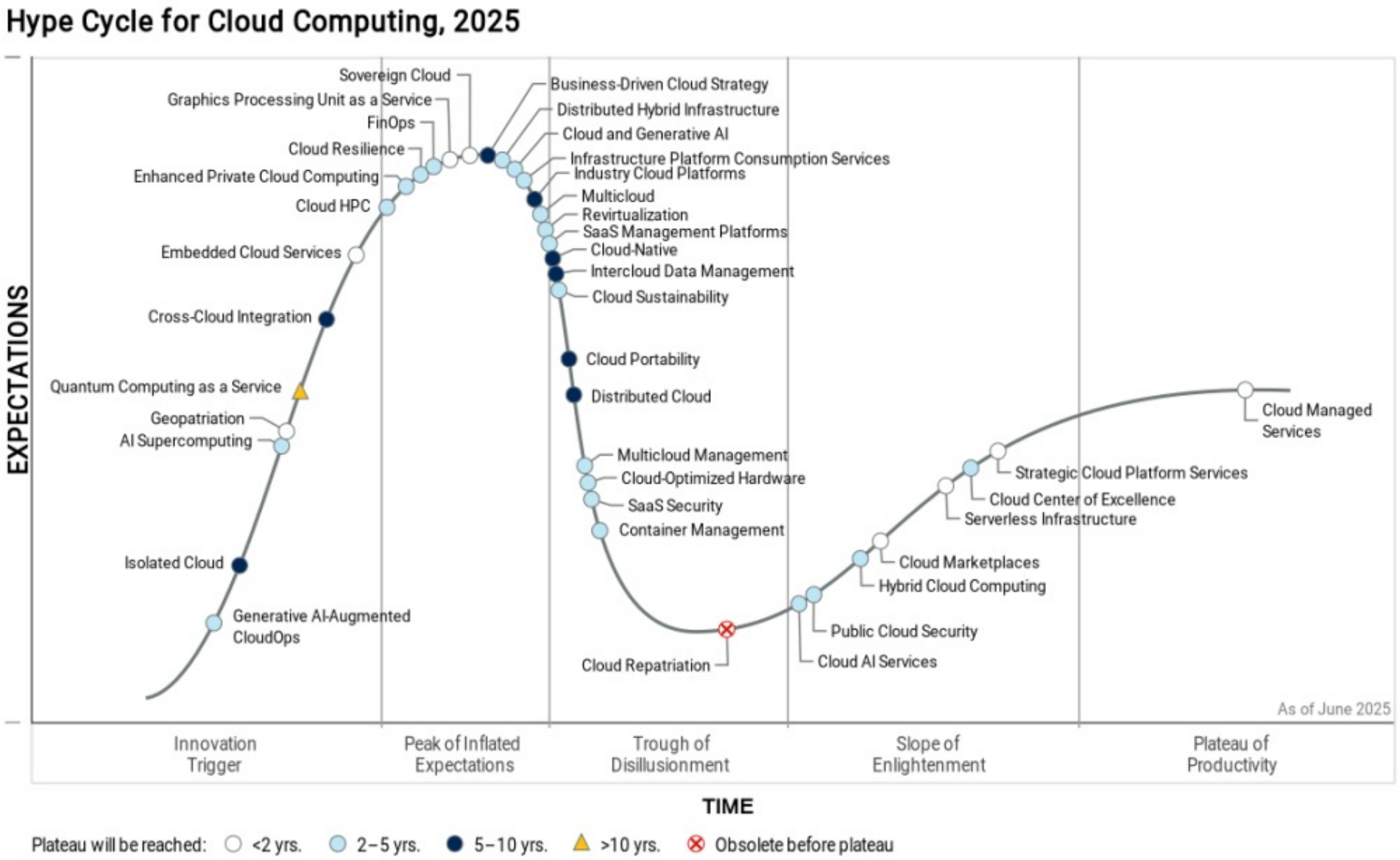
Hinweis(e): Weltweit
Quelle(n): Gartner; ID 284706

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Cloud Computing – Market Figures



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Hype Cycle for Cloud Computing, 2025
EXPECTATIONS
Sovereign Cloud
Graphics Processing Unit as a Service
FinOps
Cloud Resilience
Enhanced Private Cloud Computing
Cloud HPC
Embedded Cloud Services
Cross-Cloud Integration
Quantum Computing as a Service
Geopatriation
AI Supercomputing
Isolated Cloud
Generative AI-Augmented CloudOps
Business-Driven Cloud Strategy
Distributed Hybrid Infrastructure
Cloud and Generative AI
Infrastructure Platform Consumption Services
Industry Cloud Platforms
Multicloud
Revirtualization
SaaS Management Platforms
Cloud-Native
Intercloud Data Management
Cloud Sustainability
Cloud Portability
Distributed Cloud
Multicloud Management
Cloud-Optimized Hardware
SaaS Security
Container Management
Cloud Repatriation
Public Cloud Security
Cloud AI Services
Cloud Marketplaces
Hybrid Cloud Computing
Strategic Cloud Platform Services
Cloud Center of Excellence
Serverless Infrastructure
Cloud Managed Services
As of June 2025
Innovation Trigger
Peak of Inflated Expectations
Trough of Disillusionment
Slope of Enlightenment
Plateau of Productivity
TIME
Plateau will be reached: <2 yrs. 2–5 yrs. 5–10 yrs. >10 yrs. Obsolete before plateau

Gartner

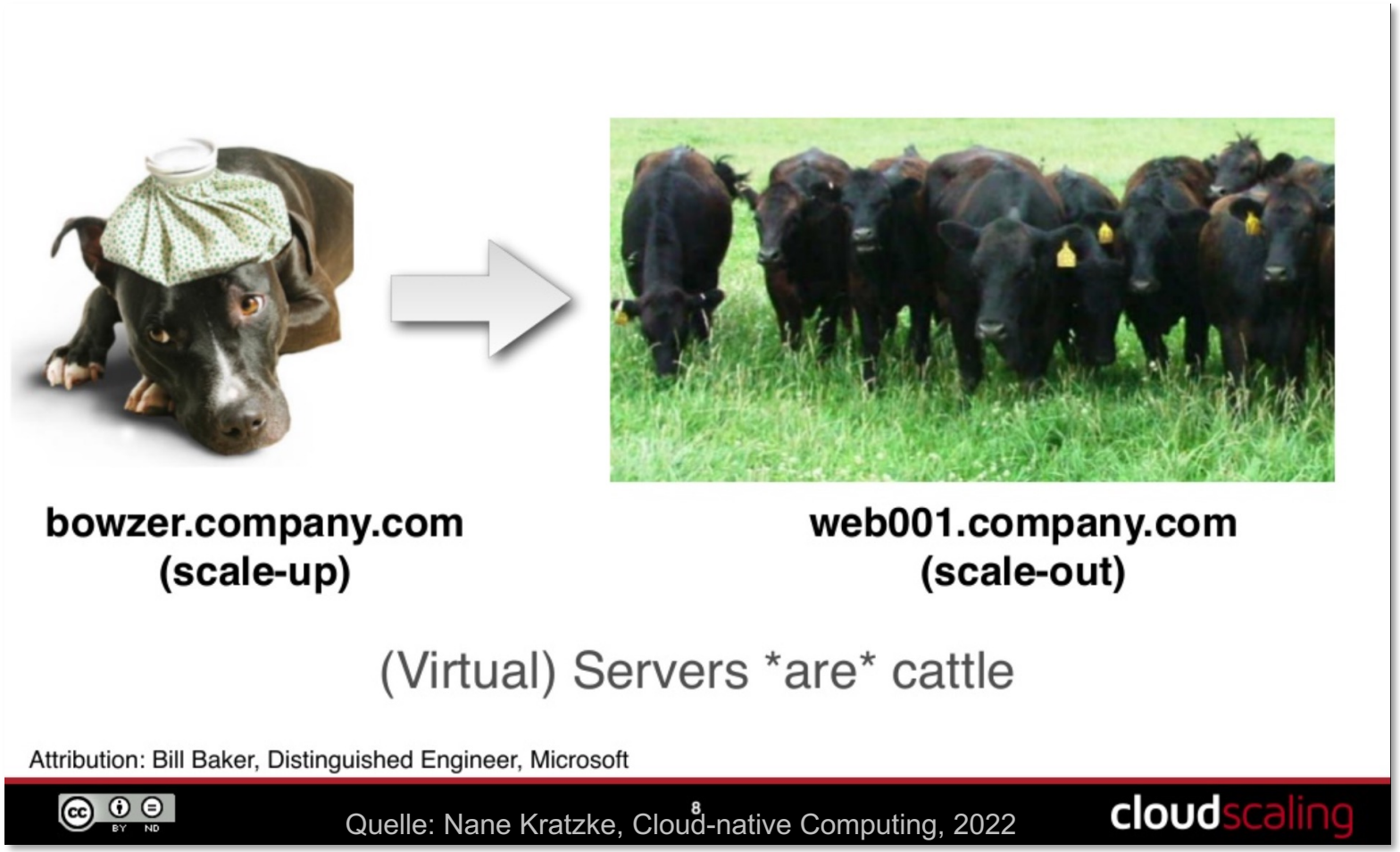
Other types of computing

- A supercomputer is often made up of thousands of processors working in parallel to perform calculations quickly. The processors are usually connected to each other by a high-speed network. Supercomputers often use specialized HW + SW.
- Cluster computing refers to the connection of multiple computers (also known as nodes) over a network so that they can function as a single (distributed) system. Unlike a single supercomputer, a cluster is often made up of off-the-shelf computers.
- In grid computing a virtual supercomputing infrastructure is formed from a collection of interconnected computers (often referred to as "nodes"). These nodes can be geographically dispersed. In contrast to cluster computing, grid computing is therefore more decentralized.
- Web or service computing refers to the creation, use, and management of web services in a distributed computing environment. It involves delivering and using software applications as services over the Internet. These "web" services can be used by other applications or services to automate complex tasks and business processes.
- Today, these "web" services are often operated on cloud platforms or infrastructure (as in NIST definition).



PETS vs. Cattle

Cloud Computing is a Mindset Change



- Alte Informatiker-Weisheit:
  - Ausschalten, Einschalten, geht wieder.
- Server sind Wegwerf-Artikel!
  - Wegschmeißen - neu machen
  - Ist schneller als Problem suchen, Problem beheben.
- Mehr Power
  - kommt durch mehr Server.

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Cloud-Native Computing

Case Study: ShareWise



ShareWise: Shared expenses made simple

A collaborative application for managing expenses within groups

ShareWise helps groups of people

- record shared expenses
- distribute costs among participants
- understand who owes whom
- record settlement payments
- close and archive completed groups

- Example use cases?



ShareWise scenario: Mallorca 2027

Four friends plan a trip together:

- **Alice** creates the group
- **Bob**, **Carol**, and **David** join
- the group records expenses during the trip
- each expense has a payer and a cost distribution
- members can record settlement payments
- the group is closed after the trip

Expense	Paid by	Amount	Participants
Hotel	Alice	€900	Everyone
Dinner	Bob	€180	Everyone
Rental car	Carol	€420	Alice, Carol, David
Museum	David	€75	Bob, Carol, David



ShareWise: Core concepts

What are the core concepts?

- Group
- Member
- Expense
- …



ShareWise: From modular monolith to cloud-native system

